

Goldman Sachs Financial Risk Analysis

Summary Report

1. Project Overview

Financial institutions process thousands of customer transactions every day. Analyzing these transactions helps identify customer behavior, account activity, financial risk, and potential anomalies.

The objective of this project was to analyze a banking transaction dataset using Python and apply data analytics techniques to understand customer transaction patterns, identify financial risks, build customer profiles, perform exploratory data analysis, and validate insights through statistical hypothesis testing.

2. Project Objectives

The main objectives of this project were to:

- Clean and prepare the banking dataset for analysis.
- Perform descriptive transactional analysis.
- Build customer profiles based on transaction behavior.
- Identify financial risks using analytical techniques.
- Perform exploratory data analysis using visualizations.
- Validate business assumptions through hypothesis testing.
- Generate business insights and recommendations based on the analysis.

3. Dataset Overview

The dataset contained **800 banking transactions** consisting of customer, account, and transaction-related information.

The dataset includes information such as:

- Customer ID
- Account ID

- Account Type
- Transaction Type
- Transaction Amount
- Account Balance
- Credit Rating
- Risk Score
- Transaction Date
- Region
- Product
- Relationship Manager

After data cleaning, the dataset was prepared for further analysis.

4. Methodology

The project was completed in six stages:

1. Data Cleaning & Formatting
2. Descriptive Transactional Analysis
3. Customer Profile Building
4. Financial Risk Identification
5. Exploratory Data Analysis
6. Hypothesis Testing

Python libraries such as **Pandas**, **NumPy**, **Matplotlib**, **Seaborn**, and **SciPy** were used throughout the project.

Task 1 – Data Cleaning & Formatting

Objective

Prepare the dataset by correcting inconsistencies and ensuring data quality before analysis.

Approach

- Converted Transaction Date into datetime format.
- Corrected negative transaction amounts.
- Corrected invalid Risk Score values.
- Verified missing values and data consistency.
- Created additional time-based columns for analysis.

Key Findings

- Dataset successfully cleaned.
- **24** negative transaction amounts were corrected.
- **35** invalid Risk Score values were corrected.
- **16** overdraft transactions were identified.
- The cleaned dataset was exported for subsequent analysis.

Insights

Data cleaning improved the reliability of the dataset and ensured that all subsequent analyses were performed on consistent and accurate data.

Task 2 – Descriptive Transactional Analysis

Objective

Analyze customer transaction patterns over time and identify account activity.

Approach

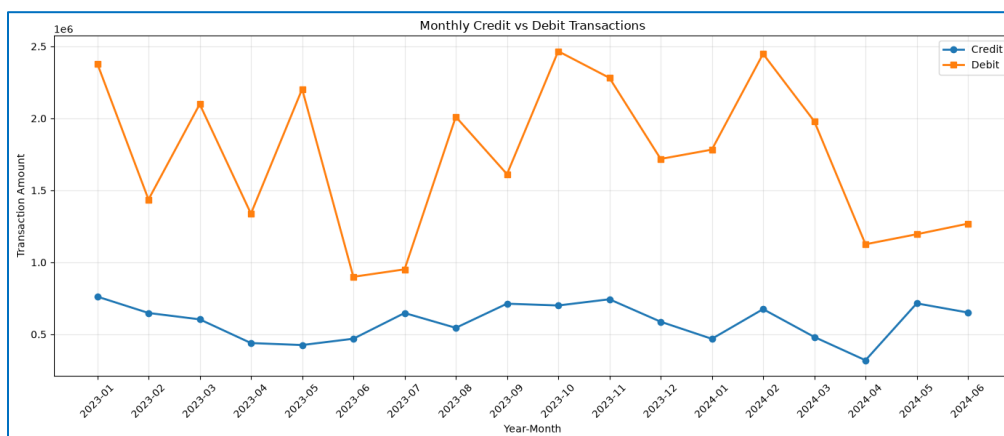
- Calculated monthly and yearly transaction summaries.
- Measured total credit and debit amounts.
- Identified top and bottom performing accounts.
- Identified dormant accounts using a 60-day inactivity criterion.

Key Findings

- Total credit and debit values were calculated for both 2023 and 2024.
- Debit transaction values exceeded credit transaction values during both years.
- **164 dormant accounts** were identified.
- Top-performing and low-performing accounts were successfully identified.

Insights

The higher debit values indicate that customers generally withdrew or spent more money than they deposited during the analysis period. Dormant accounts may require customer engagement initiatives to encourage account activity.



Task 3 – Customer Profile Building

Objective

Segment customers based on transaction behavior and account characteristics.

Approach

Customers were classified using:

- Transaction frequency
- Average account balance
- Total transaction volume
- Net inflow

Key Findings

Activity Level Distribution:

- High Activity Accounts: **73**
- Medium Activity Accounts: **76**
- Low Activity Accounts: **45** Customer Profiles:
- High Net Inflow Accounts: **49**
- High Frequency Low Balance Accounts: **34**
- Near Zero Balance Accounts: **1**

Insights

Most accounts belonged to either the High or Medium activity groups, indicating active customer engagement. A relatively small number of customers maintained low balances despite frequent transactions, suggesting accounts used primarily for regular financial activity rather than savings.

Task 4 – Financial Risk Identification

Objective

Identify accounts that exhibit potential financial risk.

Approach

Risk identification was performed using:

- Large withdrawal analysis
- Overdraft analysis
- Balance volatility (Coefficient of Variation)
- IQR anomaly detection
- Z-Score anomaly detection
- Customer risk indicators

Key Findings

- Total Overdraft Transactions: **16**
- Frequent Overdraft Accounts: **1**
- High Balance Volatility Accounts: **45**
- IQR Anomalies: **0**
- Z-Score Anomalies: **0**
- Suspicious Customers Identified: **0**

Insights

Although some accounts showed higher balance volatility, very few accounts repeatedly entered overdraft. No suspicious customers were identified based on the defined criteria, suggesting that the dataset represents relatively stable customer behavior.

Task 5 – Exploratory Data Analysis

Objective

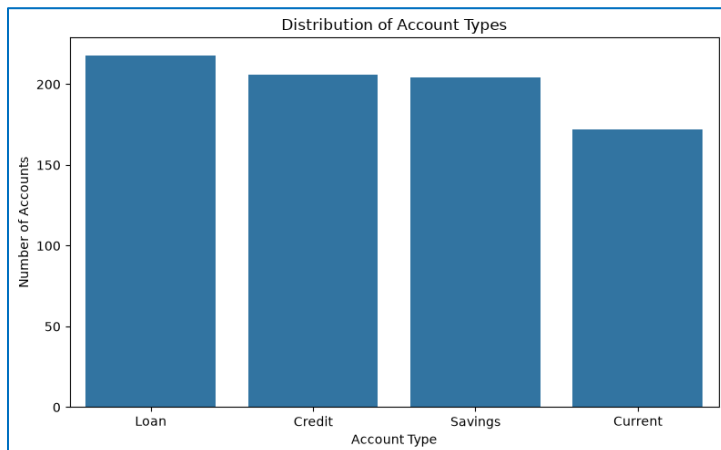
Visualize customer transaction behavior and financial characteristics.

Approach

Ten visualizations were created, including:

- Account Type Distribution
- Transaction Type Distribution
- Transaction Amount Distribution
- Monthly Transaction Trend
- Average Balance by Account Type
- Risk Score Distribution
- Credit Rating vs Risk Score
- Correlation Heatmap
- Monthly Credit vs Debit Trend
- Regional Transaction Analysis

Chart 1 – Distribution of Account Types



Purpose: This chart was created to understand how customers are distributed across different account types and determine whether any account type dominates the dataset.

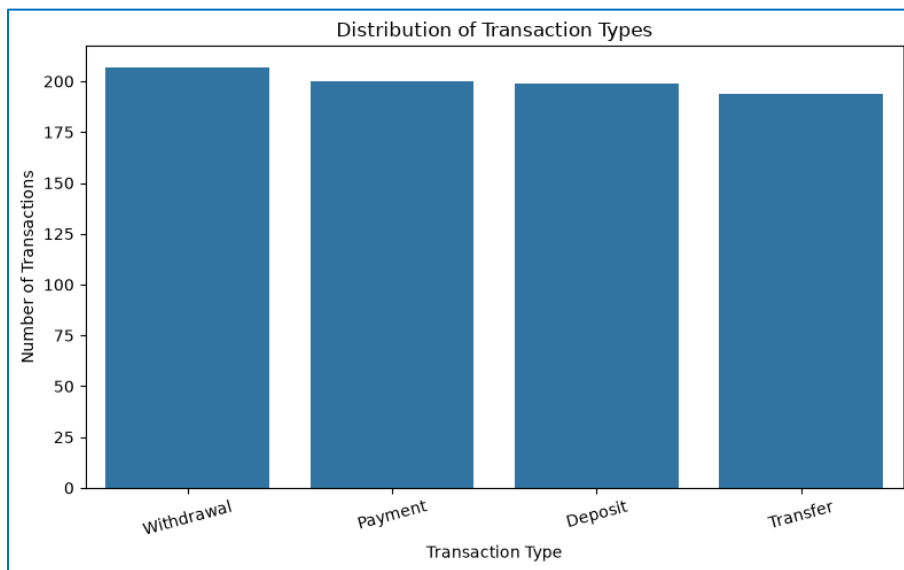
Key Findings

- Loan accounts were the most common account type (218), followed closely by Credit (206) and Savings (204) accounts.
- Current accounts had the lowest representation (172).
- The distribution is relatively balanced across all account types.

Insight

The customer base is well diversified across different account types, indicating that the dataset represents multiple banking services rather than being concentrated in a single product category.

Chart 2 – Distribution of Transaction Types



Purpose: This chart was created to analyze the frequency of different transaction types performed by customers.

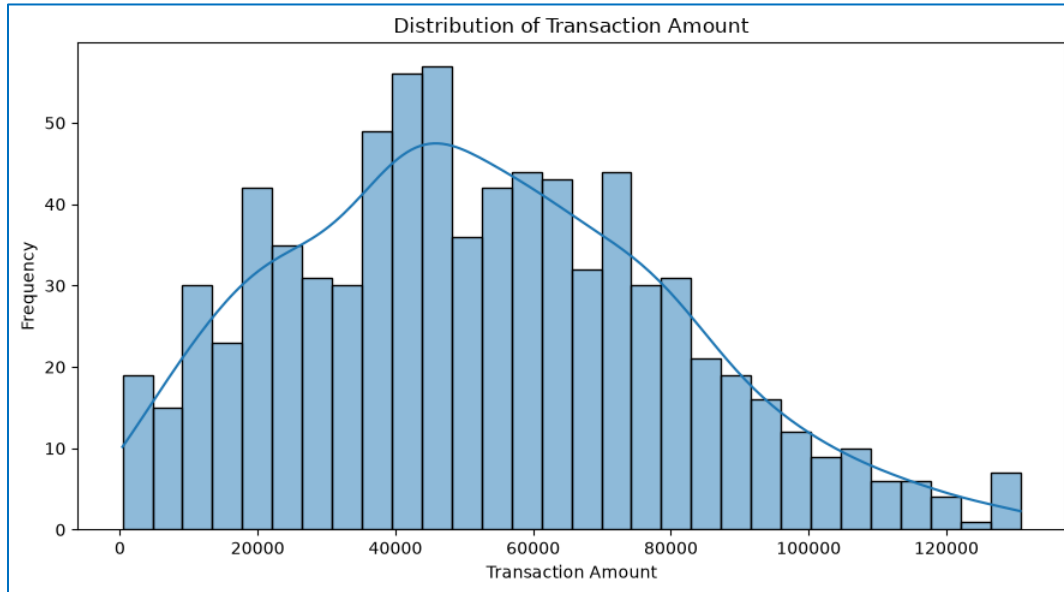
Key Findings

- Withdrawals (207), Payments (200), Deposits (199), and Transfers (194) occurred in very similar numbers.
- No transaction type overwhelmingly dominated the dataset.

Insight

Customers appear to use all major banking services with similar frequency, suggesting balanced transaction behaviour and varied banking activities.

Chart 3 – Distribution of Transaction Amount



Purpose

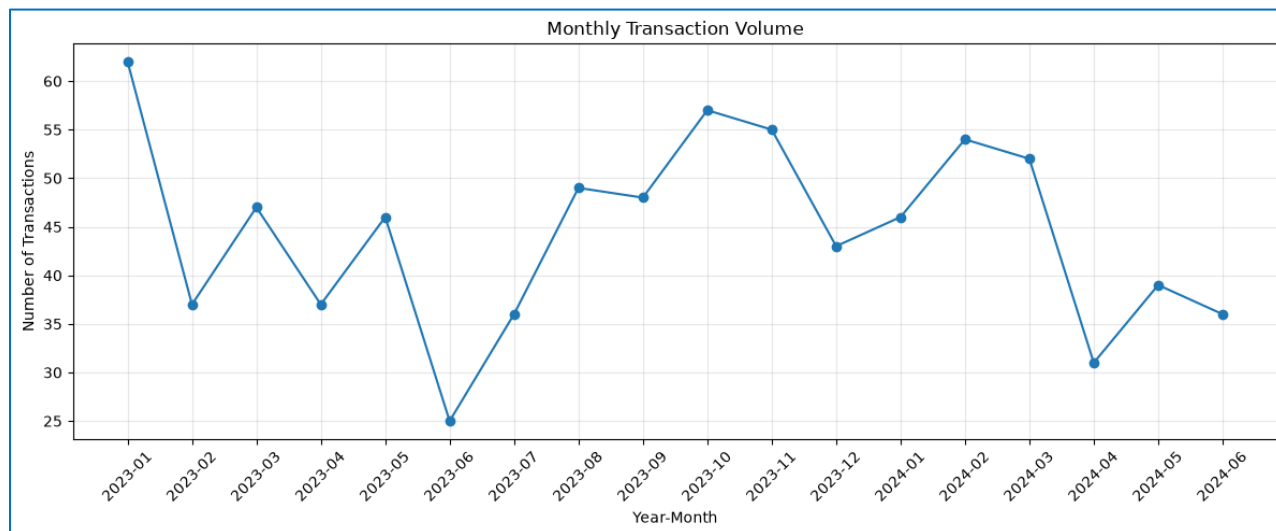
This visualization was used to examine how transaction amounts are distributed and to identify common transaction ranges or unusual values.

Key Findings

- Most transaction amounts fall between **₹40,000 and ₹70,000**.
- The distribution is slightly right-skewed, with fewer transactions exceeding **₹100,000**.
- High-value transactions are relatively uncommon.

Insight: Most customer transactions are of moderate value, while extremely large transactions occur infrequently. This indicates relatively consistent transaction behaviour across customers.

Chart 4 – Monthly Transaction Volume



Purpose

This chart was created to study transaction activity over time and identify periods of higher or lower customer engagement.

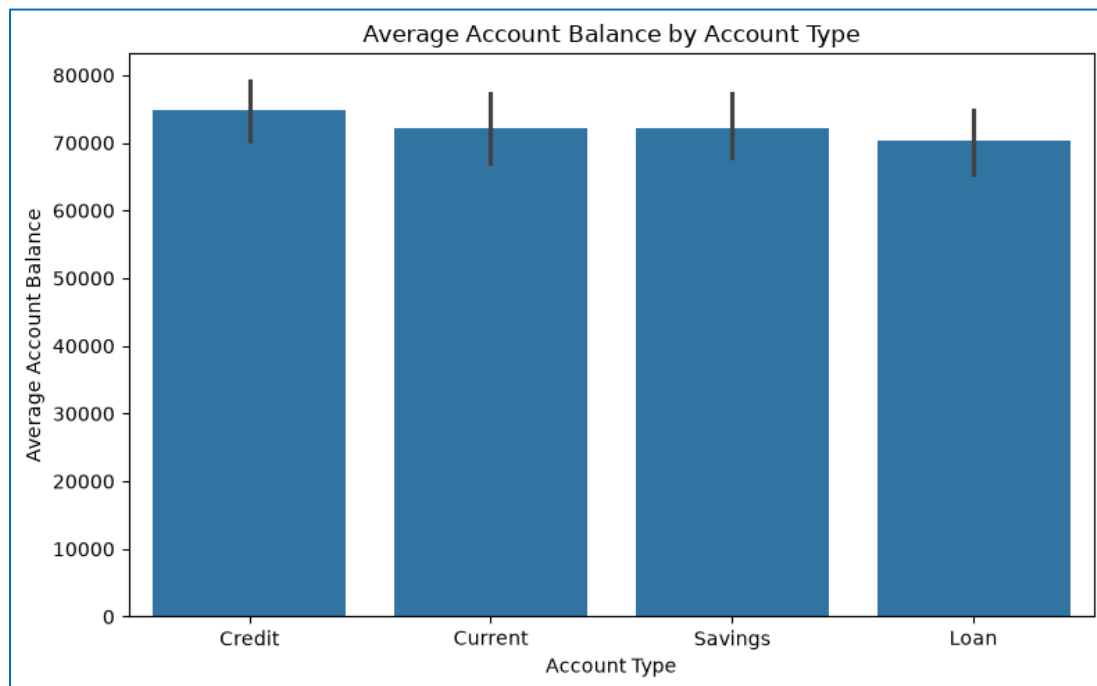
Key Findings

- Transaction volume fluctuated across different months.
- January 2023 recorded the highest number of transactions.
- June 2023 recorded the lowest transaction volume.
- Transaction activity increased again during October–November 2023 before declining in later months.

Insight

Customer transaction activity is not constant throughout the analysis period. Monitoring monthly transaction trends can help banks identify seasonal patterns and allocate operational resources more effectively.

Chart 5 – Average Account Balance by Account Type



Purpose

This chart compares the average account balance maintained across different account types.

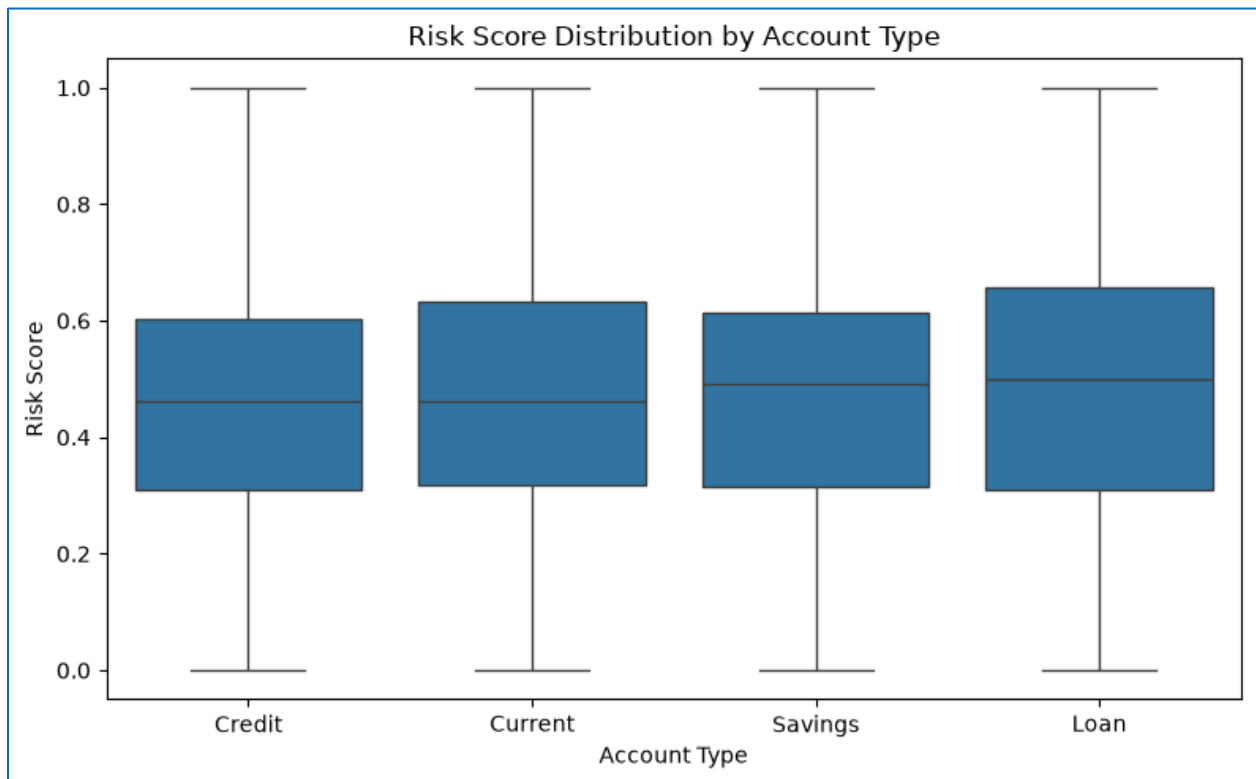
Key Findings

- Credit accounts had the highest average account balance.
- Loan accounts had the lowest average balance.
- The differences between account types were relatively small.

Insight

Customers across different account types maintain fairly similar average balances, suggesting no significant imbalance in customer financial capacity among the account categories.

Chart 6 – Risk Score Distribution by Account Type



Purpose

This visualization was created to compare the distribution of Risk Scores across different account types and identify whether any category exhibits higher financial risk.

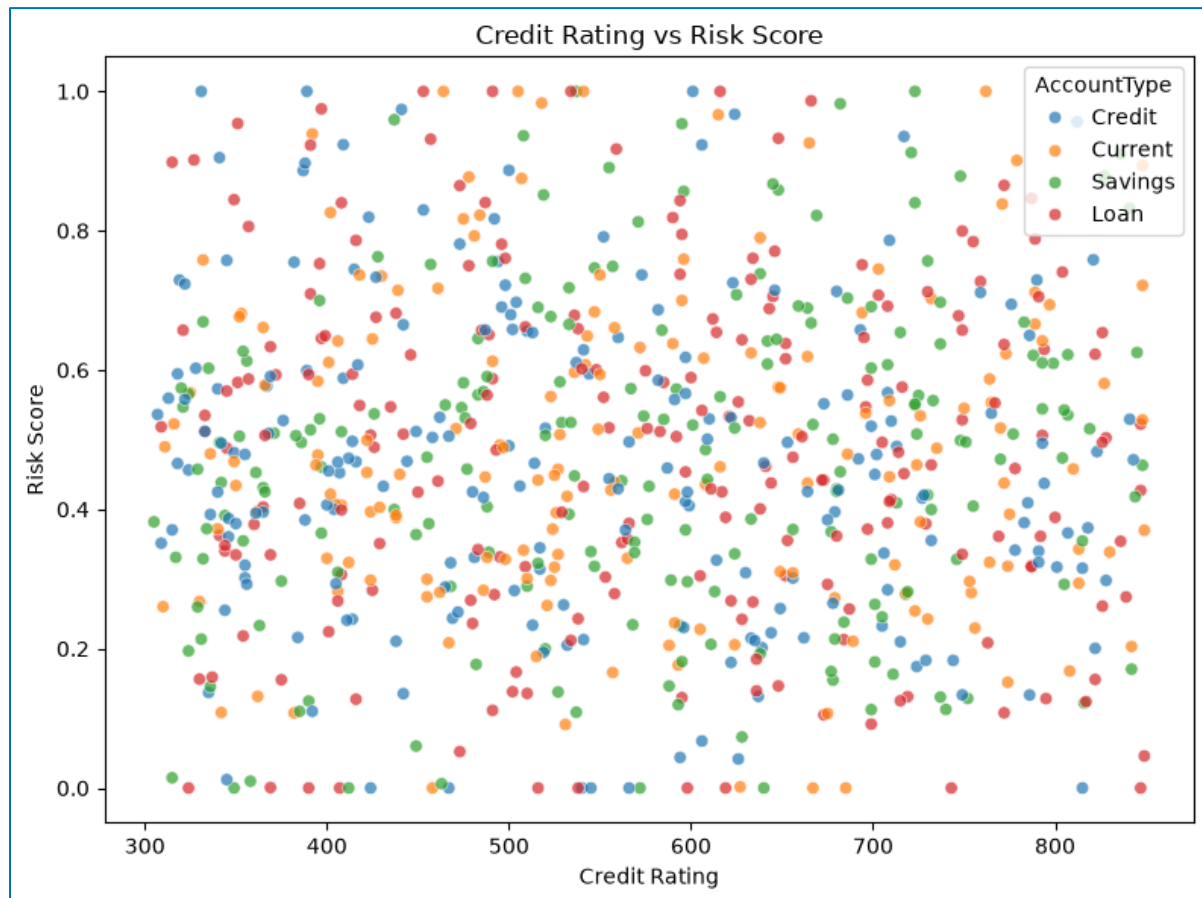
Key Findings

- Median Risk Scores were similar across all account types.
- The spread of Risk Scores was comparable for Credit, Current, Savings, and Loan accounts.
- No account type displayed significantly higher or lower overall risk.

Insight

Financial risk appears to be distributed across all account categories rather than being concentrated within a specific account type. This suggests that risk assessment should focus on individual customer behaviour instead of account type alone.

Chart 7 – Credit Rating vs Risk Score



Purpose

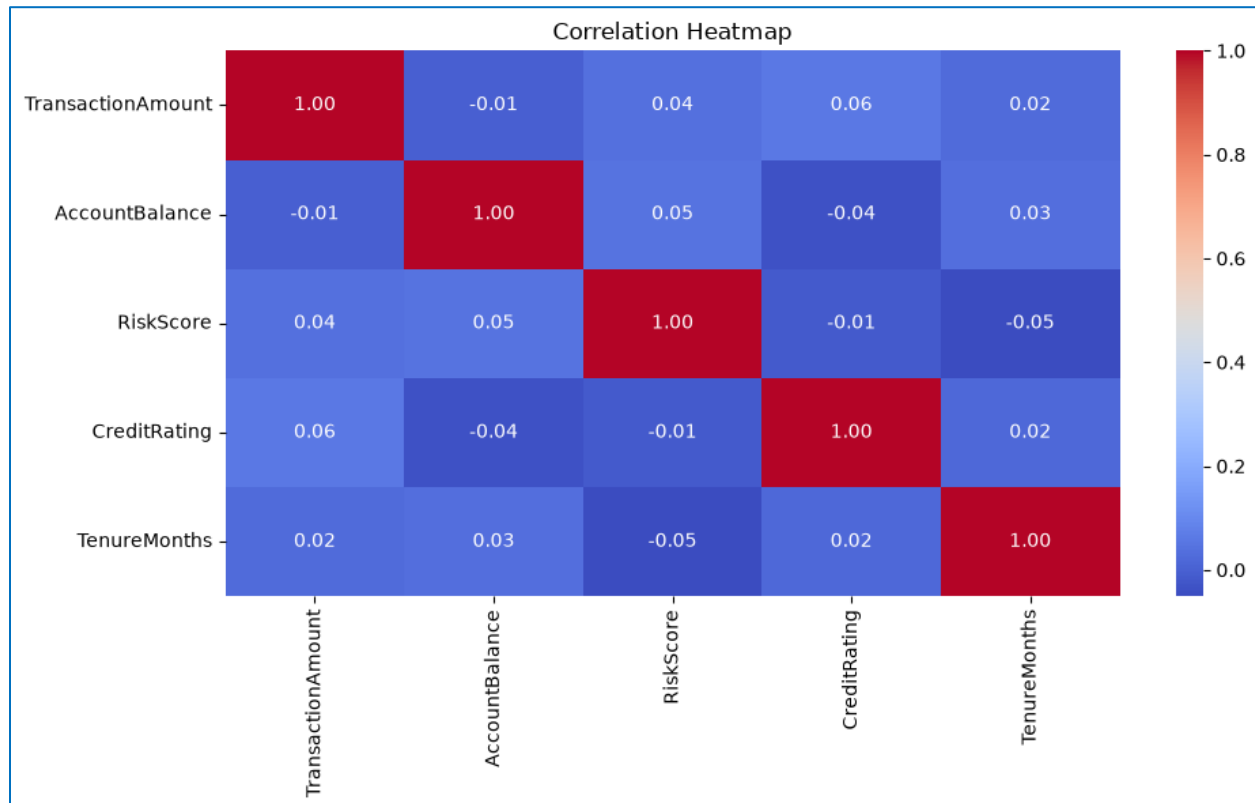
This scatter plot was created to examine whether there is a relationship between a customer's Credit Rating and Risk Score.

Key Findings

- The data points are widely scattered across the graph.
- No clear upward or downward trend is visible.
- Customers with similar Credit Ratings often have very different Risk Scores.

Insight: The analysis suggests that Credit Rating alone is not a strong predictor of Risk Score within this dataset. Other customer characteristics and financial behaviours are likely to contribute to overall financial risk. This finding also supports the hypothesis testing results, which indicate that a single financial metric is insufficient for explaining customer behaviour.

Chart 8 – Correlation Heatmap



Purpose

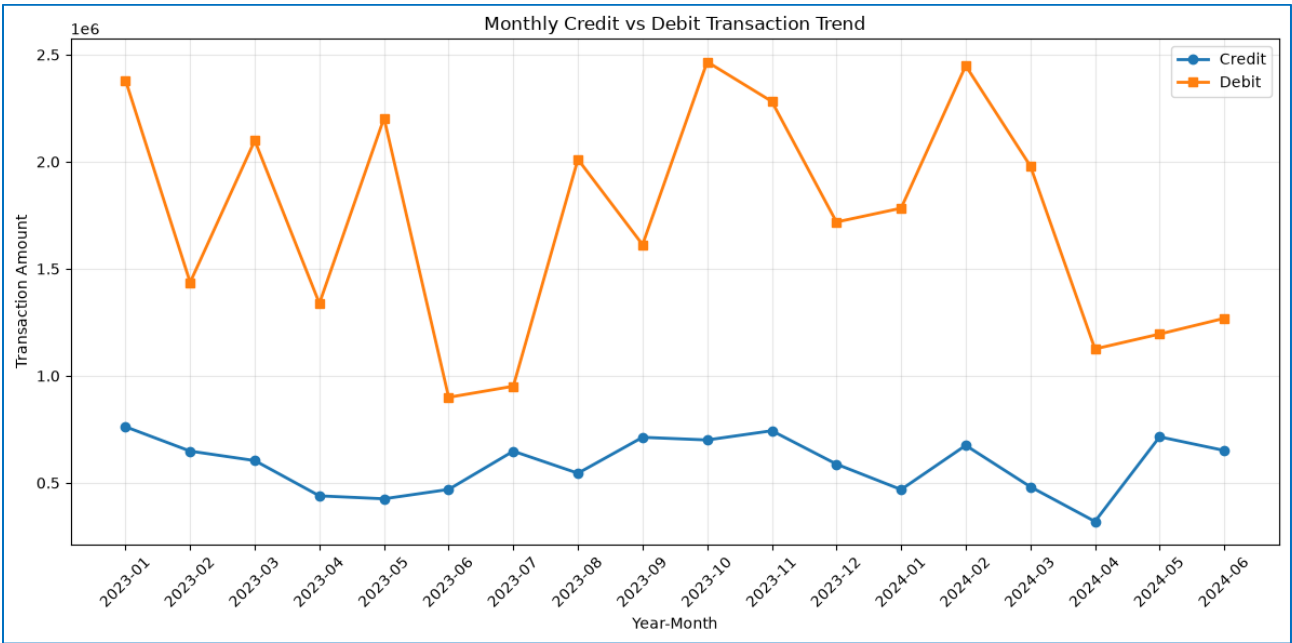
The correlation heatmap was used to measure the strength of linear relationships between the numerical variables in the dataset.

Key Findings

- All correlation coefficients are very close to zero.
- The strongest observed correlation is approximately **0.06**, indicating an extremely weak relationship.
- No pair of variables exhibits a strong positive or negative linear correlation.

Insight: The numerical variables included in this analysis are largely independent of one another. Customer financial behaviour cannot be explained by a single variable, highlighting the importance of combining multiple indicators when performing customer profiling and financial risk analysis.

Chart 9 – Monthly Credit vs Debit Transaction Trend



Purpose

This visualization compares monthly Credit and Debit transaction amounts to evaluate overall cash flow patterns and understand customer spending behaviour over time.

Key Findings

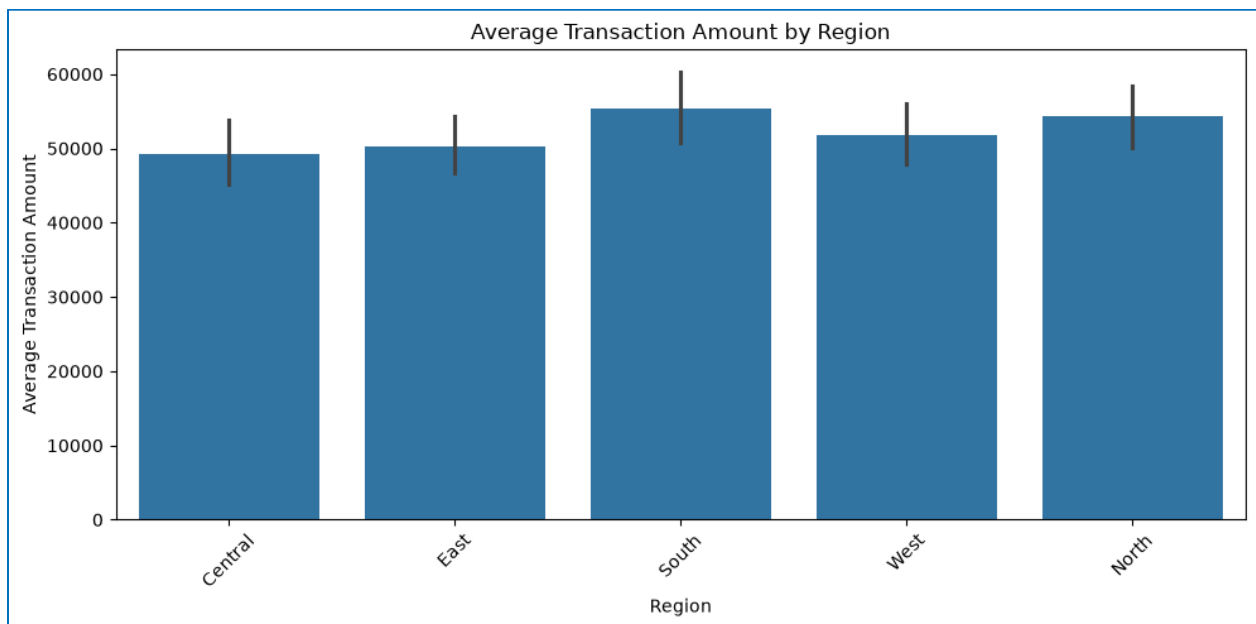
- Debit transaction amounts exceeded Credit transaction amounts in every month of the analysis period.
- Debit activity reached its highest levels around October 2023 and February 2024.
- Credit transactions remained comparatively stable with smaller month-to-month fluctuations.

Insight

The consistent dominance of Debit transactions indicates that customers generally had greater cash outflows than inflows during the analysis period. Monitoring this trend can

help banks identify changes in customer spending behaviour, evaluate liquidity patterns, and develop targeted financial products or engagement strategies.

Chart 10 – Average Transaction Amount by Region



Purpose

This chart compares the average transaction amount across different geographical regions to identify regional differences in customer transaction behaviour.

Key Findings

- South recorded the highest average transaction amount, closely followed by North.
- Central recorded the lowest average transaction amount.
- The variation across regions was relatively small.

Insight

Customer transaction values remain consistent across regions, indicating similar financial behaviour geographically. Although some regional differences exist, no region demonstrates unusually high or low transaction activity.

Task 6 – Hypothesis Testing

Objective

Determine whether high transaction volume accounts maintain significantly higher average balances than low transaction volume accounts.

Approach

- Segmented accounts using the median transaction volume.
- Performed Levene's Test.
- Conducted an Independent Samples t-Test.
- Validated results using the Mann-Whitney U Test.

Key Findings

- Levene Test p-value: **0.0168**
- Independent t-Test p-value: **0.9904**
- Mann-Whitney Test p-value: **0.4929**

The null hypothesis was **not rejected**.

Insights

The analysis did not find statistical evidence that customers with higher transaction volumes maintain higher average account balances. This indicates that transaction frequency or volume alone may not be a reliable indicator of account balance.

What Drives Customer Transaction Behavior and Financial Risk?

Based on the overall analysis, the following factors appear to influence customer transaction behavior and financial risk:

- Customers with higher transaction frequency were generally more engaged with their accounts, although frequent transactions did not necessarily correspond to higher account balances.
- Debit transaction values consistently exceeded credit values, indicating greater outgoing cash flow during the analysis period.
- Only a small number of accounts experienced repeated overdrafts, suggesting that persistent financial distress was uncommon.
- Balance volatility varied across customers, with 45 accounts showing relatively higher fluctuations, making them suitable candidates for closer monitoring.
- The absence of significant anomalies and suspicious customers indicates that the dataset reflects largely normal transaction behavior under the defined analytical criteria.

Recommendations

Based on the findings, the following recommendations are suggested:

- Monitor customers with high balance volatility more closely to identify emerging financial risks.
- Engage dormant account holders through targeted campaigns to encourage account activity.
- Continue monitoring accounts with repeated overdrafts and frequent large withdrawals to reduce financial risk.
- Combine transaction behavior with additional customer attributes rather than relying solely on transaction volume when evaluating customer value.
- Perform periodic anomaly detection as part of routine financial monitoring, even when current results indicate stable customer behavior.

Conclusion

This project successfully applied data analytics techniques to analyze banking transaction data and evaluate customer behavior and financial risk.

The analysis included data cleaning, descriptive analytics, customer segmentation, financial risk assessment, exploratory data analysis, and hypothesis testing. The findings demonstrated that while customer transaction patterns varied across accounts, the dataset showed relatively stable financial behavior with limited evidence of significant anomalies or suspicious activity.

Overall, the project illustrates how data analytics can support customer profiling, financial monitoring, and informed business decision-making within the banking sector.